

# Course milestone M8: Robustness Audit

## HONR 46400 · Evidence-Driven Research

### What to submit

One file, `lastname_m08.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M08**. Brightspace carries the deadline.

This milestone presents **Book Milestone 8: Your robustness audit** (checkpoint, version 1) of the course book, EDR|AI.

### What this course adds

#### Your second mentor meeting, requested

The second of the course's two mentor meetings is booked now, well ahead of the poster work it exists to serve. Email your instructor again, from your Purdue address, asking for a meeting inside the Round 02 window your course platform posts.

The email carries the same three things the first request carried: at least three concrete slots inside the window, your project in two sentences as it now stands, and the one decision you most want pressure-tested. Add one line this time naming what has changed since Round 01. Save the sent email as a PDF and carry it as a section of your milestone document.

**What Round 02 is for.** By then your result exists and has been audited, and the poster is the next thing you owe the conference. So the meeting works on the claim your evidence can carry, the board that will carry it, and the questions a visitor will ask you in front of it.

Send this request early. The Round 02 window sits inside the busiest stretch of the conference block, and a request that arrives late gets you no meeting at all.

### Milestone 8: Your robustness audit

#### Open the milestone workbook in Colab

**This chapter closes Studio 8: Stress-test and adjudicate.** Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

**What this milestone produces.** A pre-listed robustness grid, negative tests with their assumptions stated, diagnostics, an adversarial-review record, and your adjudication of what survived.

### What you bring

You also carry forward the last milestone's artifact: **Milestone 7: Your first reproducible analysis**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 24: Robustness and Sensitivity: your pre-listed robustness grid with its full results, direction and size, plus the named null check.
- Lesson 25: Diagnostics and Negative Tests: your negative test and diagnostic, each with the prediction you wrote before running it.
- Lesson 26: AI as Adversarial Reviewer: your commissioned adversarial review with every flag adjudicated against the data, confirmed or refuted.
- Lesson 27: Recognizing False Confidence: your confidence audit: the not-yet-recomputed list, where belief in the headline number comes from, and your boundary sentence.

## The practice

1. Pre-list your checks before you run any of them, and commit to reporting all of them.
2. Run the robustness grid and report a span within each commensurable panel, never across panels.
3. Run the null check that your design licenses, and name it correctly.
4. Commission at least one adversarial review, and add a human reviewer when one is available. Neither review is independent verification: confirm or refute every flag with a data check.
5. Write the adjudication: what survived, what did not, and what you still cannot rule out.

## A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

## The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Report the checks that hurt your finding as fully as the ones that helped.
- **Evidence, provenance, and reproducibility.** A flag is real when a check confirms it, not when the reviewer sounds certain.
- **AI activity, verification, and human decisions.** An AI reviewer is another critique, not independent verification.
- **Uncertainty, claim boundary, and revision history.** Specification spread measures your choices; it is not an uncertainty interval.

## Where this milestone sits

You arrived carrying Milestone 7: Your first reproducible analysis. Milestone 9 writes down only the claims this audit licenses, at the strength it licenses them. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

## What sends you back

A finding that does not survive its own pre-listed checks returns to Studio 5 or 7, not to the writing studio.

## How this milestone is assessed

Each row scores 0, 1, or 2. **10 points total.**

| # | Criterion                                                                 | 0                                                                   | 1                                                                              | 2                                                                       |
|---|---------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1 | The milestone artifact exists in full, specific to your own project       | absent, or generic text that would fit any project                  | present but incomplete, or specific in some parts and generic in others        | complete and unmistakably about your project                            |
| 2 | The milestone is written as a dated, numbered version with its reason     | no version, or a version with no reason given                       | versioned, but the reason is decorative rather than an account of what changed | versioned with a reason a reader could use to reconstruct your thinking |
| 3 | Every judgment in the artifact is defended by you, not asserted by a tool | conclusions appear with no reasoning, or reproduce a tool's wording | reasoning present but thin where the decision was hardest                      | each judgment carries a reason you could defend under questioning       |
| 4 | The four rails are carried in the form this studio requires               | one or more rails absent where this studio required them            | all rails present but one is nominal                                           | every rail addressed in the form this studio requires                   |
| 5 | The milestone's own product is present and defensible: A                  | the milestone's own product is missing                              | present but would not survive a careful question                               | present and defensible as stated                                        |

| # | Criterion                                                                                                                                                   | 0 | 1 | 2 |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
|   | pre-listed robustness grid, negative tests with their assumptions stated, diagnostics, an adversarial-review record, and your adjudication of what survived |   |   |   |

**! Blocking gate**

No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

**Your workbook**

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.